

PROBLEM STATEMENT #63 | Sustainability & Green Tech

Community Solar Credit Allocation Manager

A clean energy sharing portal ingesting rooftop solar generation metrics, facilitating peer-to-peer solar credit transfers within a neighborhood microgrid, and tracking monthly utility billing offsets.

Name : SACHIN B R

SRN : PES1UG25CS839

DELIVERABLE ONE

FR-001

Req ID: FR-001

Type: Functional

Description: The system shall record solar generation units (kWh) from smart meter feeds and allow prosumers to transfer surplus credits to neighbor accounts.

Priority: High

Acceptance Criteria: Pass — Valid solar generation data is recorded and surplus credits can be transferred to a neighbor account. Fail — A transfer exceeding the available credit balance is rejected.

Rationale: This is the core functionality of the Community Solar Credit Allocation Manager.

FR-002

Req ID: FR-002

Type: Functional

Description: The system shall allow Prosumer Residents to view their available solar credit balance.

Priority: High

Acceptance Criteria: Pass — The system displays the current available solar credit balance to the Prosumer Resident. Fail — The system does not display the balance or displays an incorrect balance.

Rationale: Prosumer Residents need to know their available credits before transferring them to a neighbor.

FR-002

Req ID: FR-002

Type: Functional

Description: The system shall allow Prosumer Residents to view their available solar credit balance.

Priority: High

Acceptance Criteria: Pass — The system displays the current available solar credit balance to the Prosumer Resident. Fail — The system does not display the balance or displays an incorrect balance.

Rationale: Prosumer Residents need to know their available credits before transferring them to a neighbor.

FR-004

Req ID: FR-004

Type: Functional

Description: The system shall update the solar credit balances of both the Prosumer Resident and the neighbor after a successful credit transfer.

Priority: High

Acceptance Criteria: Pass — The transferred credits are deducted from the prosumer's balance and added to the neighbor's balance. Fail — Either account is not updated correctly.

Rationale: Ensures that solar credits are accurately transferred between the two accounts.

FR-005

Req ID: FR-005

Type: Functional

Description: The system shall track monthly utility billing offsets resulting from applied solar credits.

Priority: Medium

Acceptance Criteria: Pass — The system displays the monthly utility billing offset based on the solar credits applied. Fail — The billing offset is missing or incorrect.

Rationale: Allows the system to track how solar credits reduce monthly utility bills.

NFR-001

Req ID: NFR-001

Type: Non-Functional — Performance & Security

Description: The smart meter telemetry processing pipeline must validate and store hourly generation data for 500 households with 99.9% uptime.

Priority: High

Acceptance Criteria: Pass — The system validates and stores hourly generation data for 500 households while maintaining 99.9% uptime. Fail — The system fails to meet the 500-household or 99.9% uptime requirement.

Rationale: Ensures reliable processing and storage of solar generation data at the required scale. Copy of 63_SE_Lab1_SE_Problem_Statements (1).pdfPDF

NFR-002

Req ID: NFR-002

Type: Non-Functional — Security

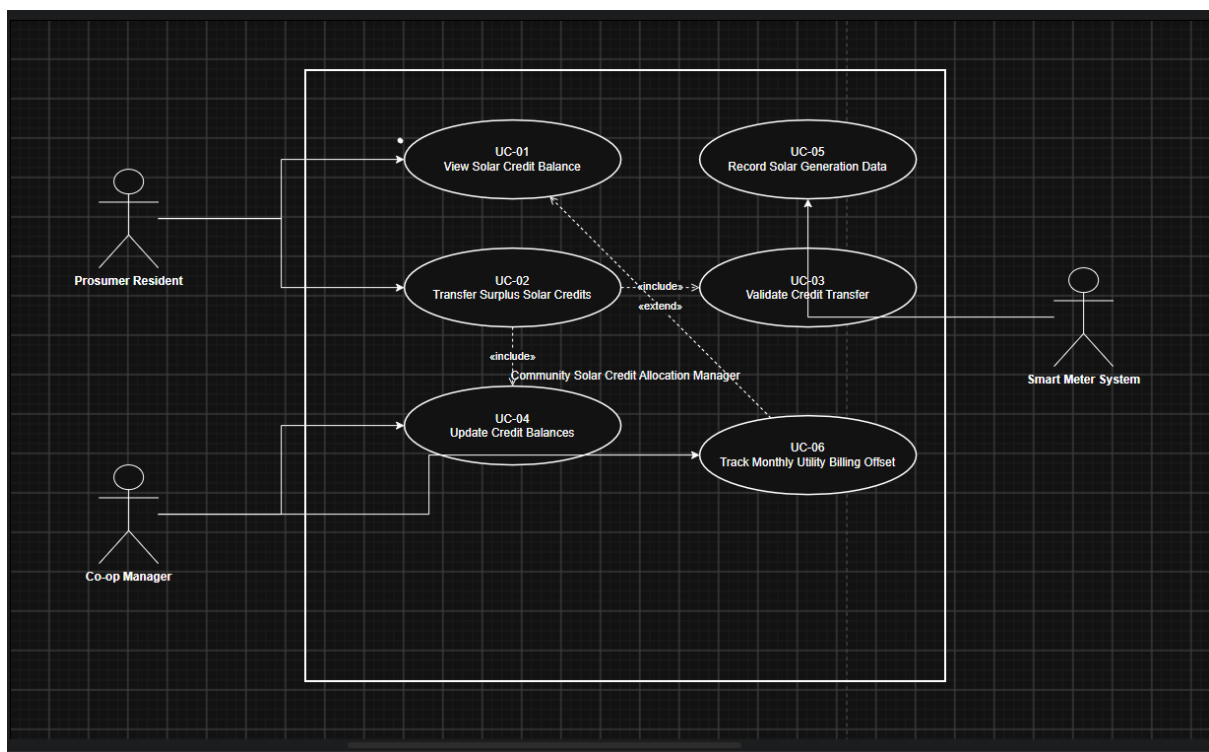
Description: The system shall ensure that only authenticated Prosumer Residents can initiate solar credit transfers from their accounts.

Priority: High

Acceptance Criteria: Pass — Only an authenticated Prosumer Resident can initiate a credit transfer. Fail — An unauthenticated or unauthorized user can initiate a transfer.

Rationale: Protects solar credits from unauthorized transfers.

DELIVERABLE TWO



DELIVERABLE THREE

Use-Case Flow Specification

Preconditions

1. The Prosumer Resident is registered and logged into the system.
2. The Prosumer Resident has available surplus solar credits.
3. The neighbor's account exists in the system.

Postconditions

- 1.The transferred credits are deducted from the Prosumer Resident's account.
- 2.The transferred credits are added to the neighbor's account.
- 3.The credit transfer transaction is recorded in the system.
- 4.The transferred credits are reflected in the neighbor's billing account.

Main Success Scenario

1. The Prosumer Resident selects "Transfer Surplus Solar Credits."
2. The system displays the available solar credit balance.
3. The Prosumer Resident enters the neighbor's account and the amount of credits to transfer.
4. The system validates the neighbor account and transfer amount.
5. The system checks that the Prosumer Resident has sufficient available credits.
6. The Prosumer Resident confirms the transfer.
7. The system deducts the transferred credits from the Prosumer Resident's balance.
8. The system adds the transferred credits to the neighbor's account.
9. The system records the credit transfer transaction.
10. The system displays a successful transfer confirmation.
11. The use case ends successfully

Insufficient Solar Credits

1. The system determines that the requested transfer amount exceeds the Prosumer Resident's available credit balance.
2. The system displays an "**Insufficient solar credits**" message.
3. The system does not process the transfer.
4. The Prosumer Resident can enter a lower amount or cancel the transfer.
5. If the Prosumer Resident cancels, the use case ends without changing either account.

This alternate flow directly follows the problem statement's requirement that a transfer exceeding the available balance should fail.

